Sampling Techniques

The Imperative of Sampling

Cost & Efficiency

Reduces the financial and temporal drain of research, making large-scale data collection viable.

Scalable Generalization

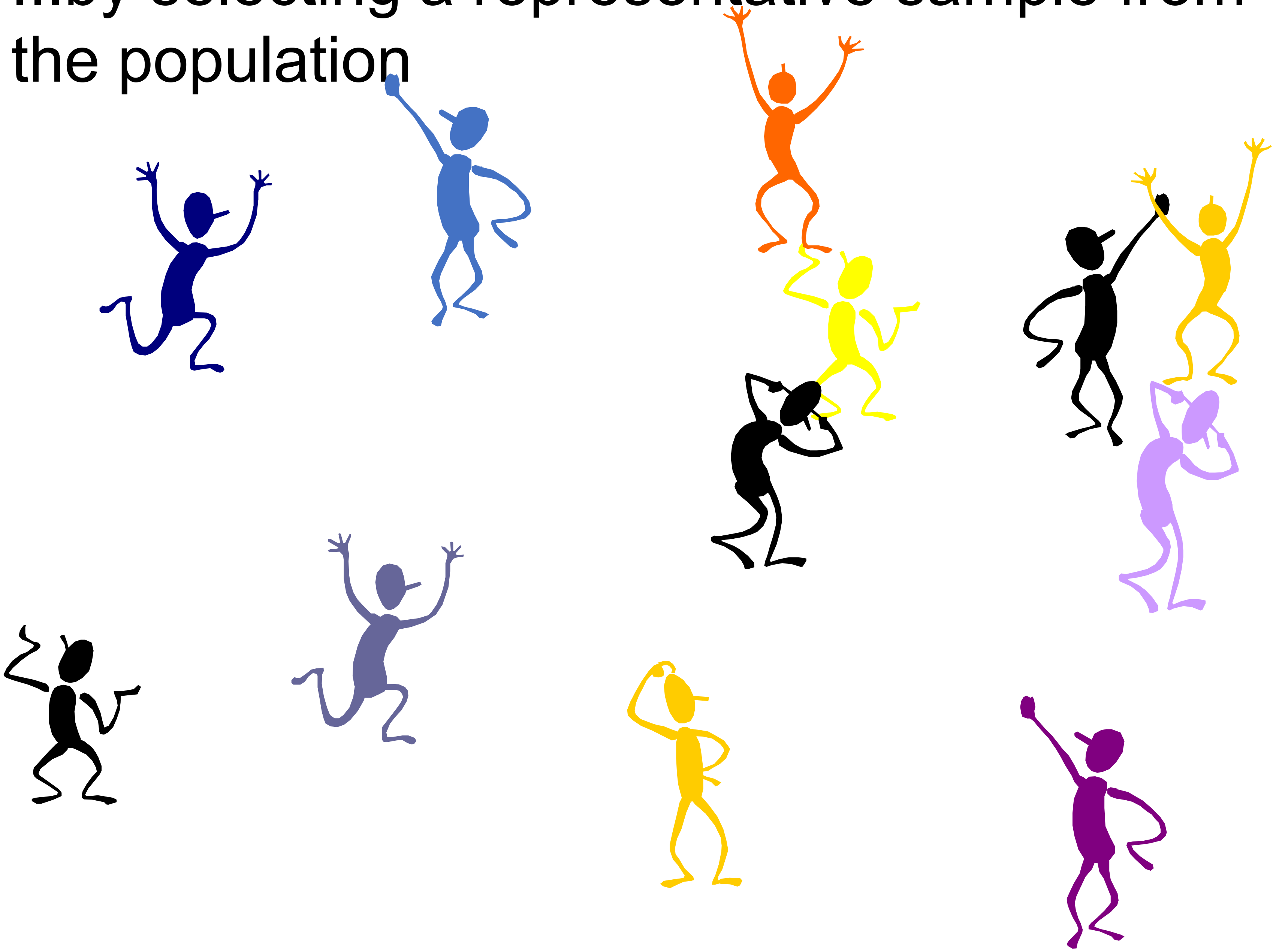
Allows researchers to infer accurate population-level truths by analyzing a mathematically representative cross-section.

Non-Destructive Analysis

Essential for industrial testing where analyzing the entire population would mean destroying the entire inventory.

Population inferences can be made...

...by selecting a representative sample from the population



The Master Variable: Choosing a Paradigm

	QUALITATIVE PARADIGM	QUANTITATIVE PARADIGM
Research Objective	In-depth exploration and understanding of phenomena.	Statistical representation and large-scale generalization.
Key Metric for Size	Attainment of theoretical saturation.	Calculated allowable sampling error and risk.
Typical Size Range	Small: 15–50 participants. Focus groups: 6-8 per group.	Large: Determined by precision formulas and population variance.
Selection Logic	Judgement, expertise, and active, targeted selection.	Probability sampling and rigid mathematical design parameters.

Qualitative Selection Framework

Maximum Variation

Captures a broad range of contrasting subjects to understand diverse attitudes and beliefs.

Deviant (Outliers)

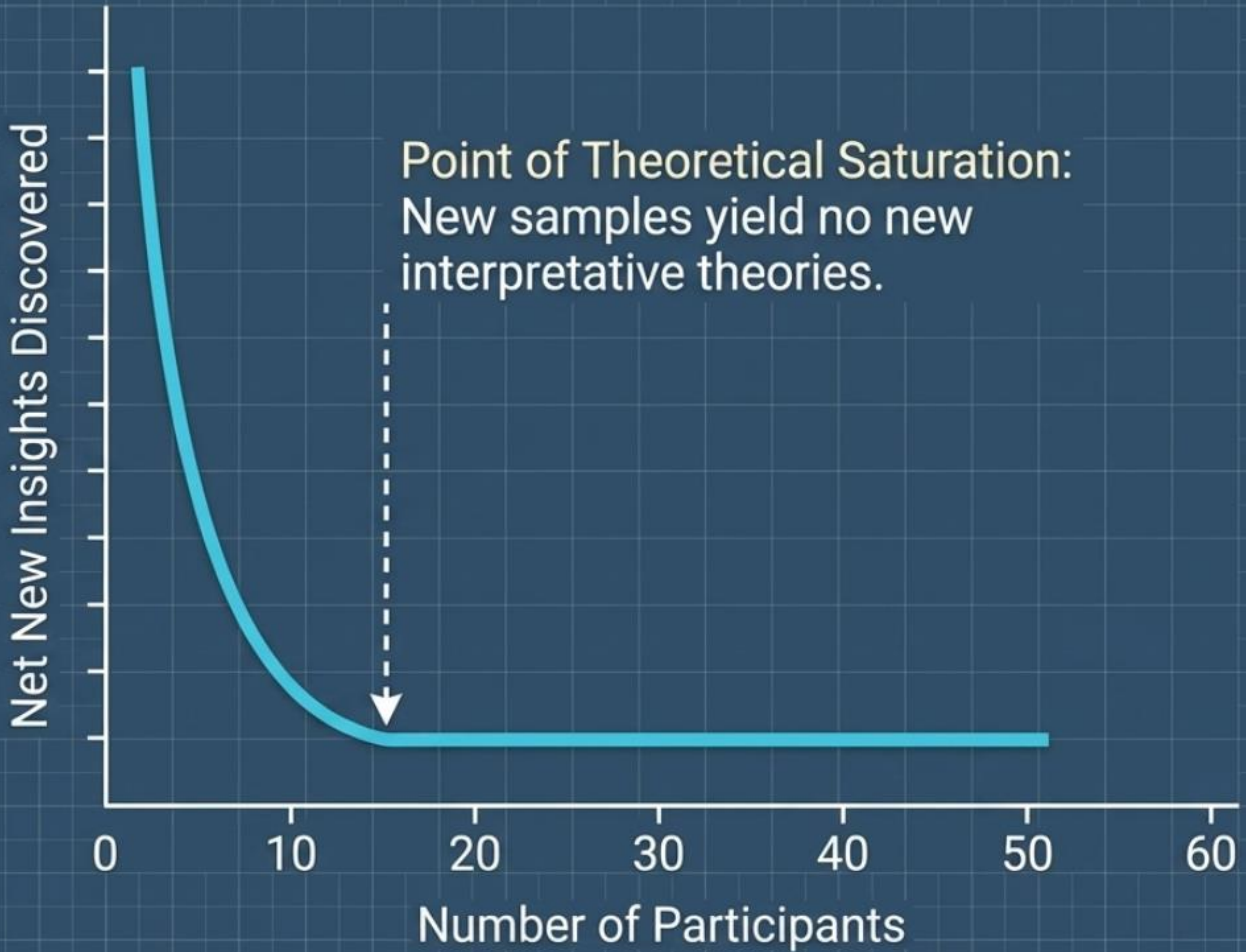
Focuses on highly unusual or extreme cases to test the boundaries of an emerging theory.

Critical Case

Targets specific subjects whose unique experiences make them the structural linchpin for understanding the phenomena.

Key Informant

Selects subjects possessing highly specialized knowledge or oversight of the research area.



Hard Rules

- Minimum participants: 15.
- Maximum ceiling: Under 50.
- Focus Group Discussions (FGDs): 6-8 participants per group, minimum of 2 FGDs per population group.

Four Approaches to Quantitative Sizing

1. **The Census**
   Surveying the entire population.

2. **Imitation**
   Duplicating a similar, peer-reviewed study.

3. **Published Tables**
   Using standardized lookup reference grids.

4. **Mathematical Formulas**
   Calculating precise values (e.g., Cochran, Fisher).

Approaches 1 & 2: Census and Imitation

The Census

- **Rules:** Strictly for small populations (200 or less).
- **Benefit:** Completely eliminates sampling error. Fixed costs (design, frame) remain identical whether surveying 50 or 200.

The Imitation Method

- **Rules:** Adopting the sample size of a similar discipline study (must strictly cite reference).
- **Critical Risk:** Blindly replicating another researcher's inherited methodological errors without conducting a rigorous procedural review.

Approach 3: Published Tables (The Lookup Method)

Population Size	+/- 10% Precision	+/- 7% Precision	+/- 5% Precision
100	51	67	81
200	67	101	134
300	76	121	172
400	81	135	201

Key Insight: Demanding higher precision (moving left from +/- 10% to +/- 5%) forces a massive, non-linear exponential increase in required sample size across the same population.

Approach 4: The Mathematical Formula

n_0: The target calculated sample size.

Z^2: The Confidence Level. The abscissa of the normal curve (Standard benchmark: Z = **1.96** for a **95%** level of confidence).

$$n_0 = \frac{Z^2 pq}{e^2}$$

q: Calculated simply as (1 - p).

e^2: The desired level of statistical precision.

p: The **estimated proportion** of an attribute present in the target population.

Note: Cochran (1963) is ideal for large proportions. Other formulas like Fisher or Mugenda exist for specific variant designs.

Real-World Adjustments & Complex Designs

Base Sample (n0)

The theoretical minimum calculated via formula or published table.

+ 10% Buffer

Added universally to compensate for uncontactable individuals in the field.

+ 30% Buffer

Added specifically for studies with high non-response rates (e.g., self-administered questionnaires).

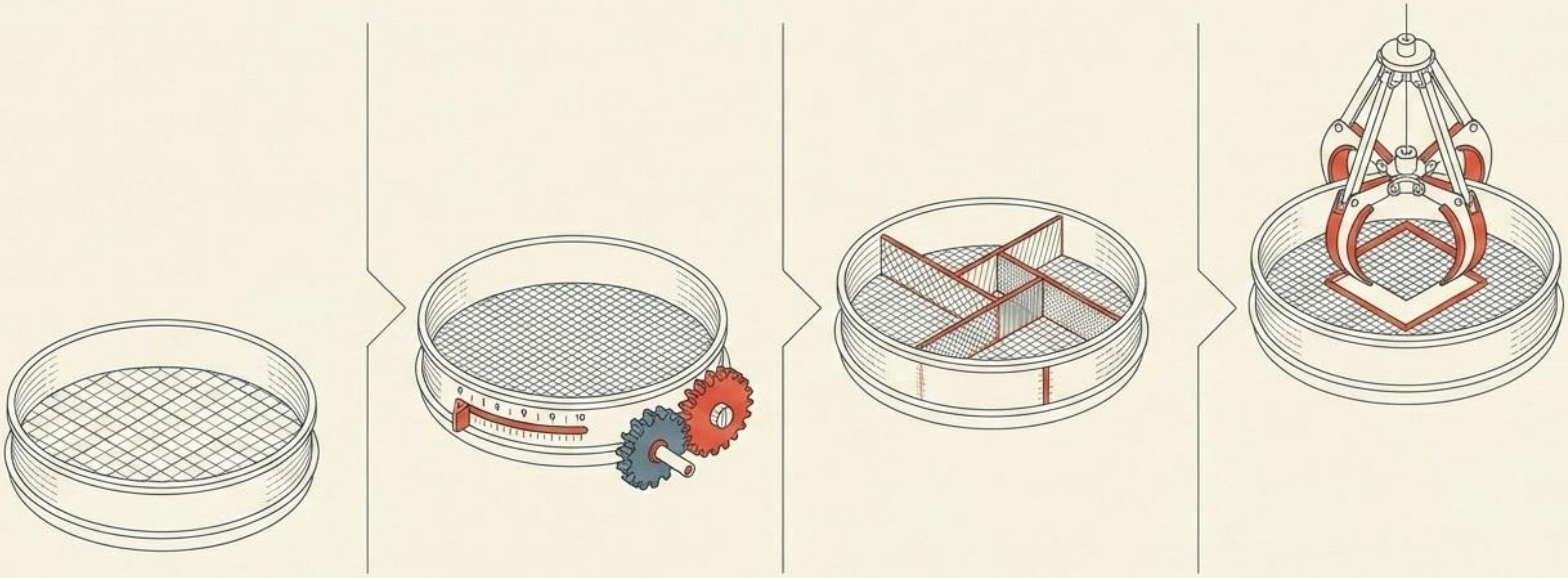
Final Target Output

The actual gross number of surveys or invites the researcher must deploy.

Complex Designs: Simple random sampling formulas are merely the baseline. Complex designs (case-control, stratified, or cluster sampling) require further mathematical adjustments to account for the variances of sub-populations.

Types of Random Sampling

Four Tools for Data Filtration



Simple

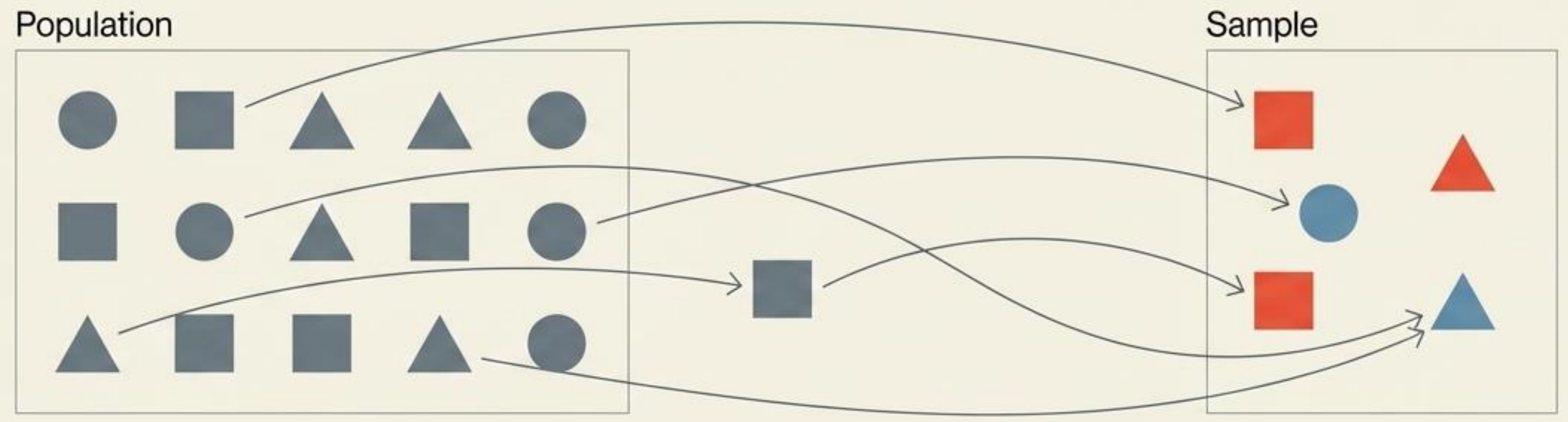
Systematic

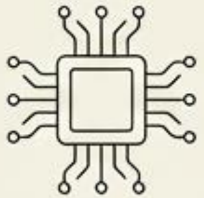
Stratified

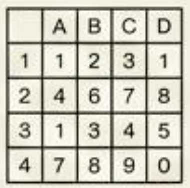
Cluster

Simple Random Sampling

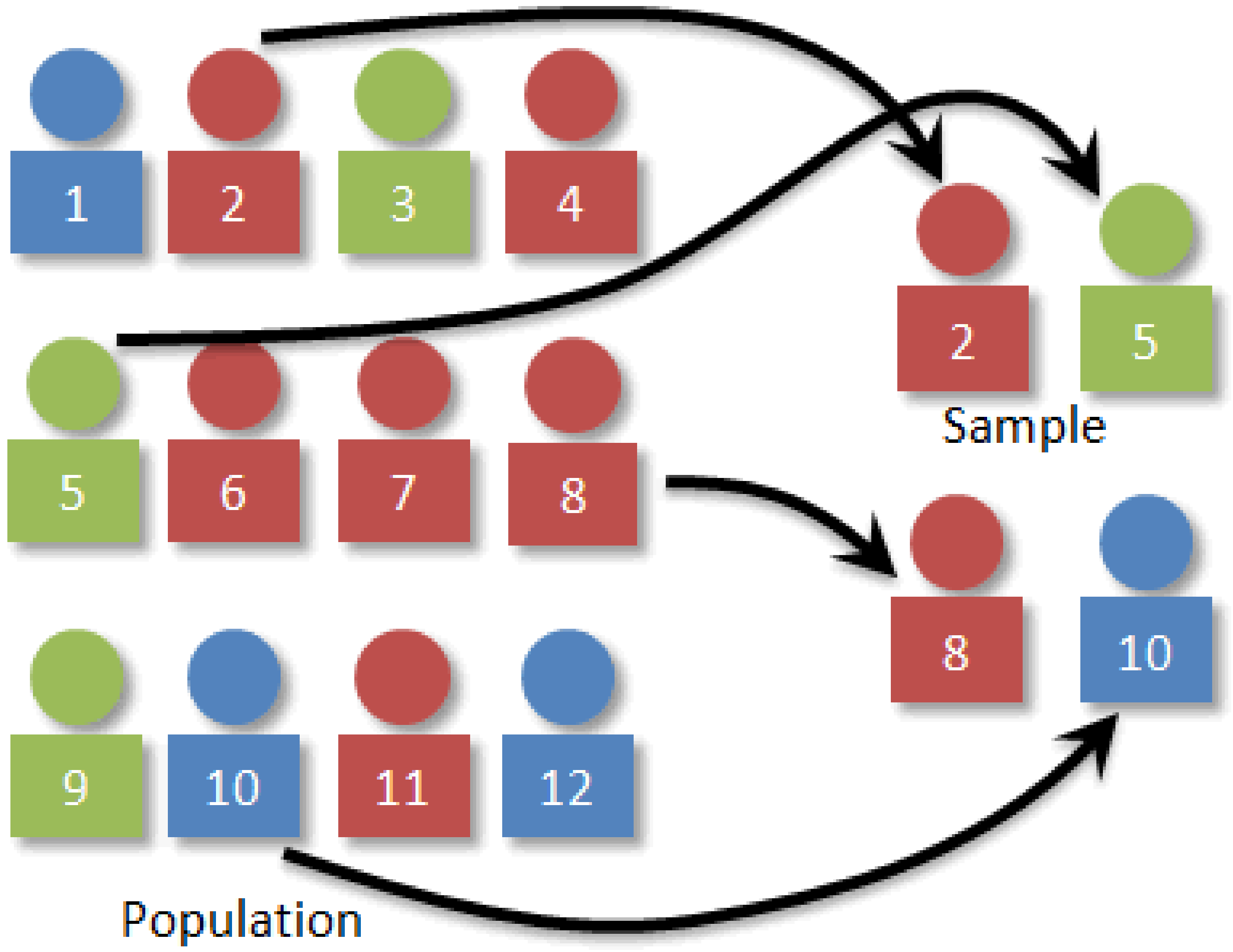
The Baseline of Pure Probability







1	Every member of the population has an equal chance of being chosen.	2	Every combination of N members has an equal chance of being chosen.

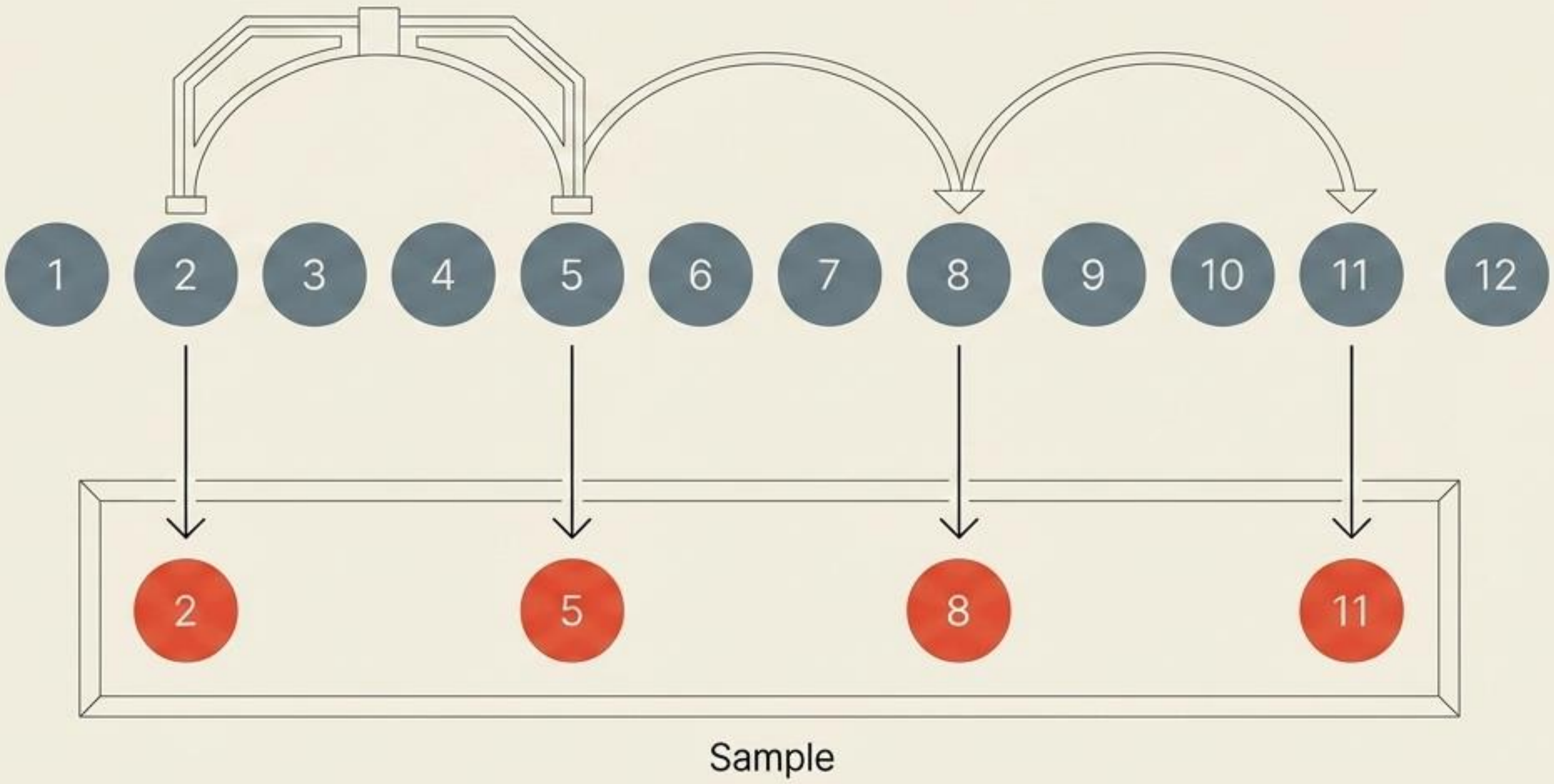

1
2
3
4
5
6
7
8
9
10
11
12
Population
2
5
8
10
Sample

Systematic Random Sampling

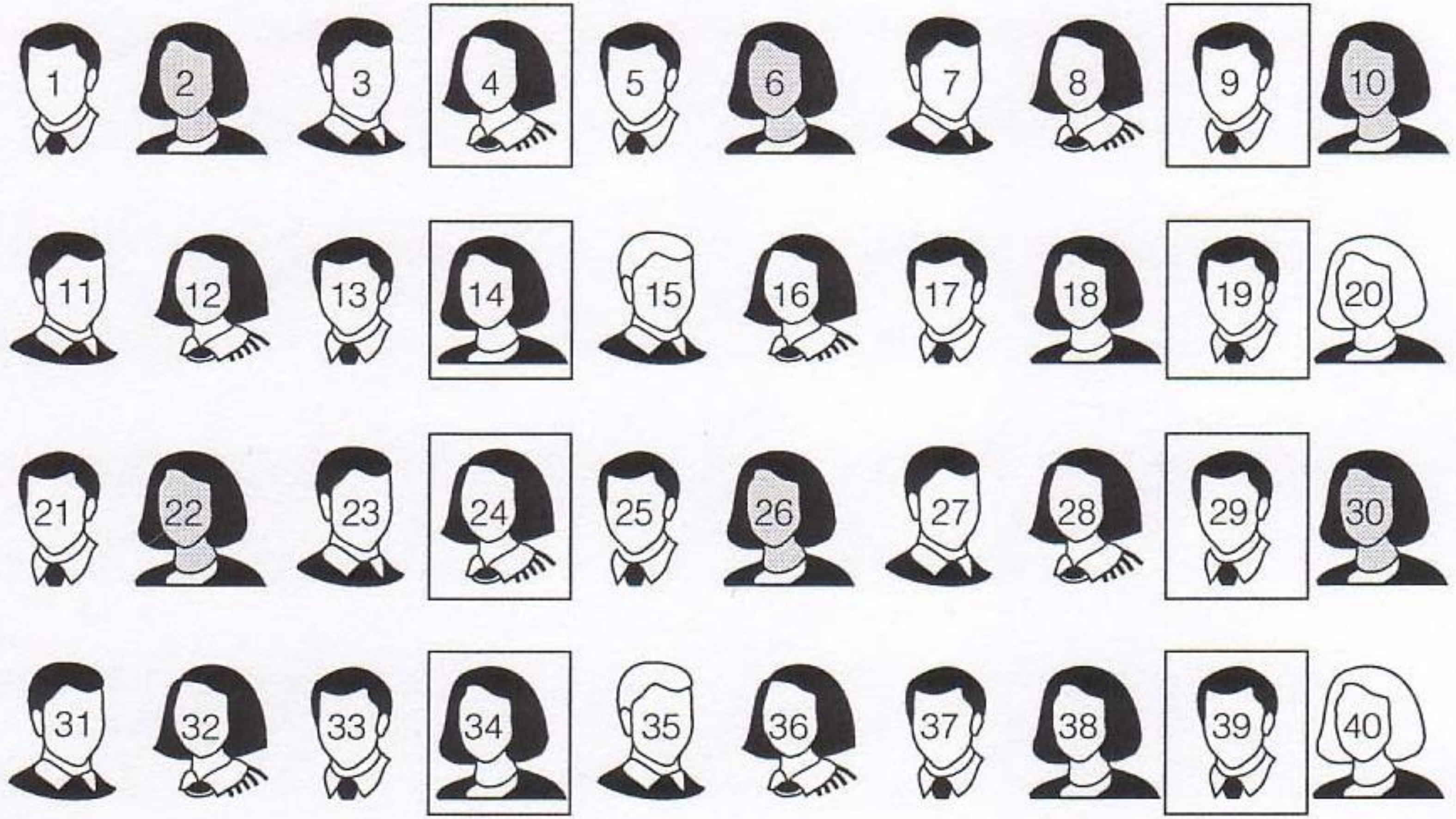
Adding Mechanical Efficiency with Systematic Sampling

Step 1:
Randomly select the first member from among the first K members.

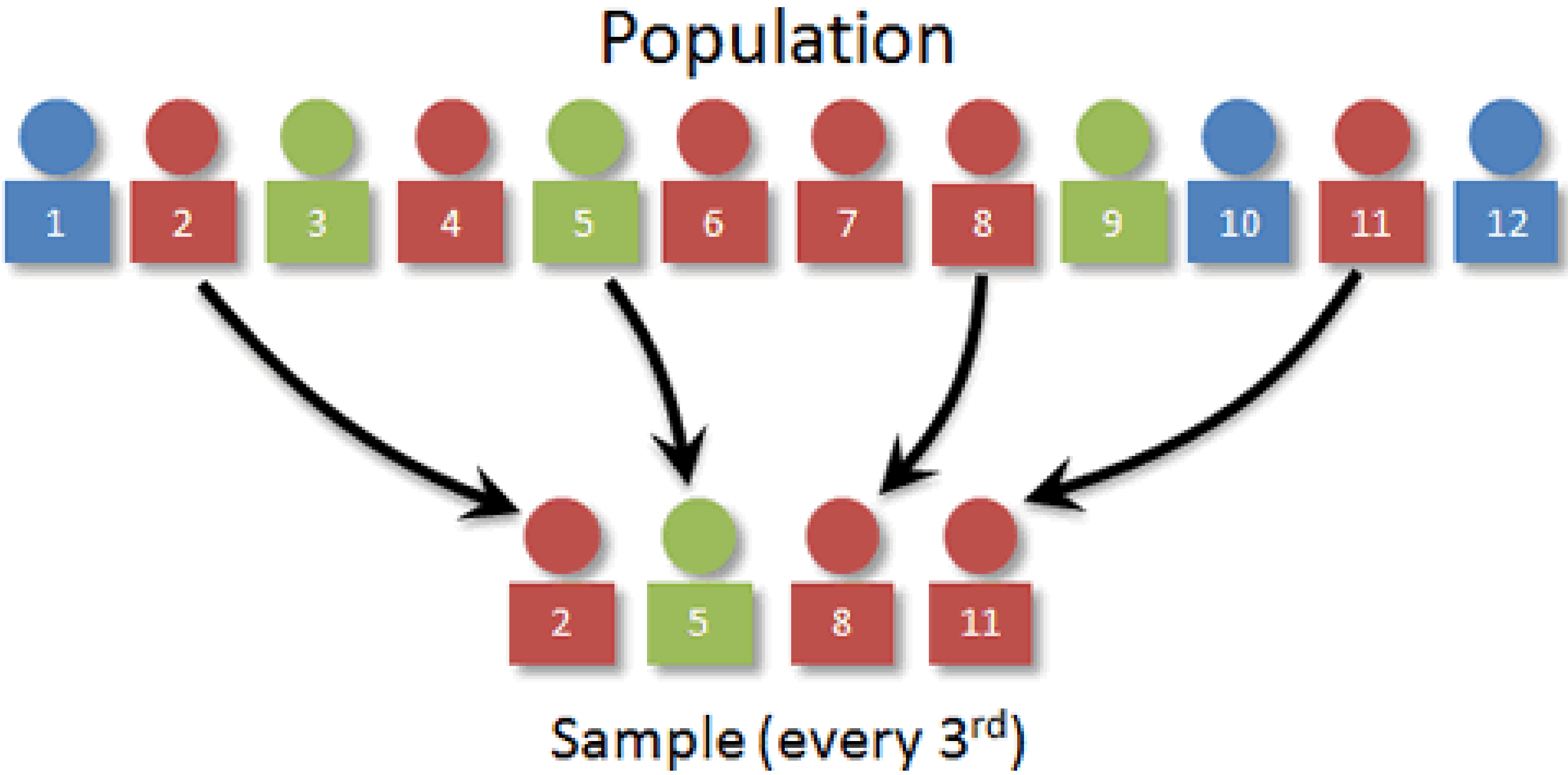
Step 2:
Automatically select every Kth member thereafter.



From a population of 40 students, let's select a systematic random sample of 8 students. Our skip interval will be 5 (40 ÷ 8 = 5). Using a random number table, we choose a number between 1 and 5. Let's say we choose 4. We then start with student 4 and pick every 5th student:

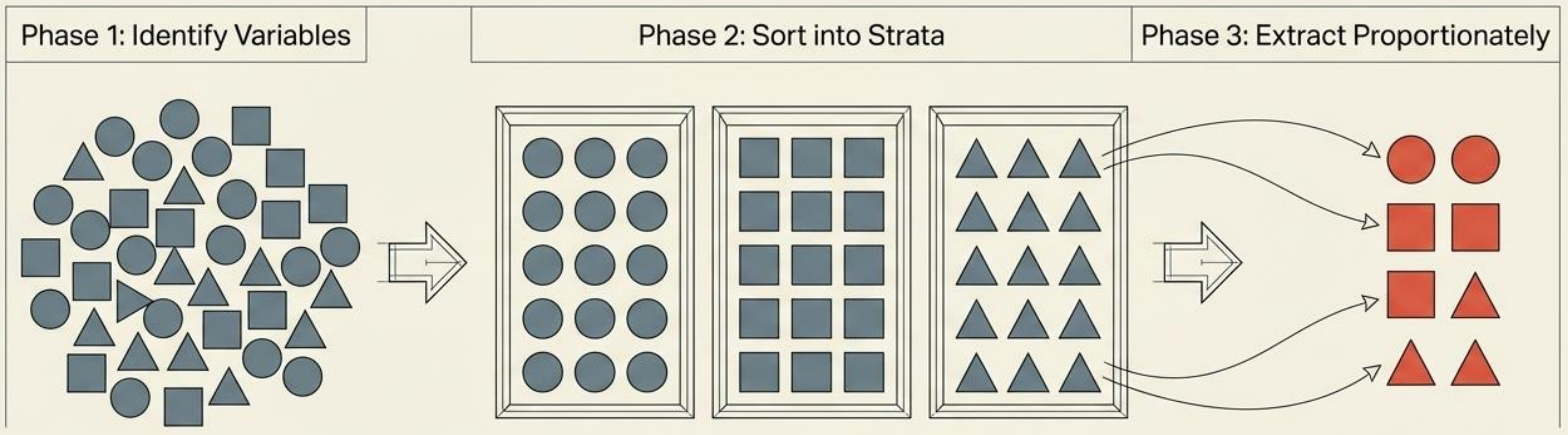


Our trip to the random number table could have just as easily given us a 1 or a 5, so all the students do have a chance to end up in our sample.


Population
1
2
3
4
5
6
7
8
9
10
11
12
2
5
8
11
Sample (every 3rd)

Stratified Random Sampling

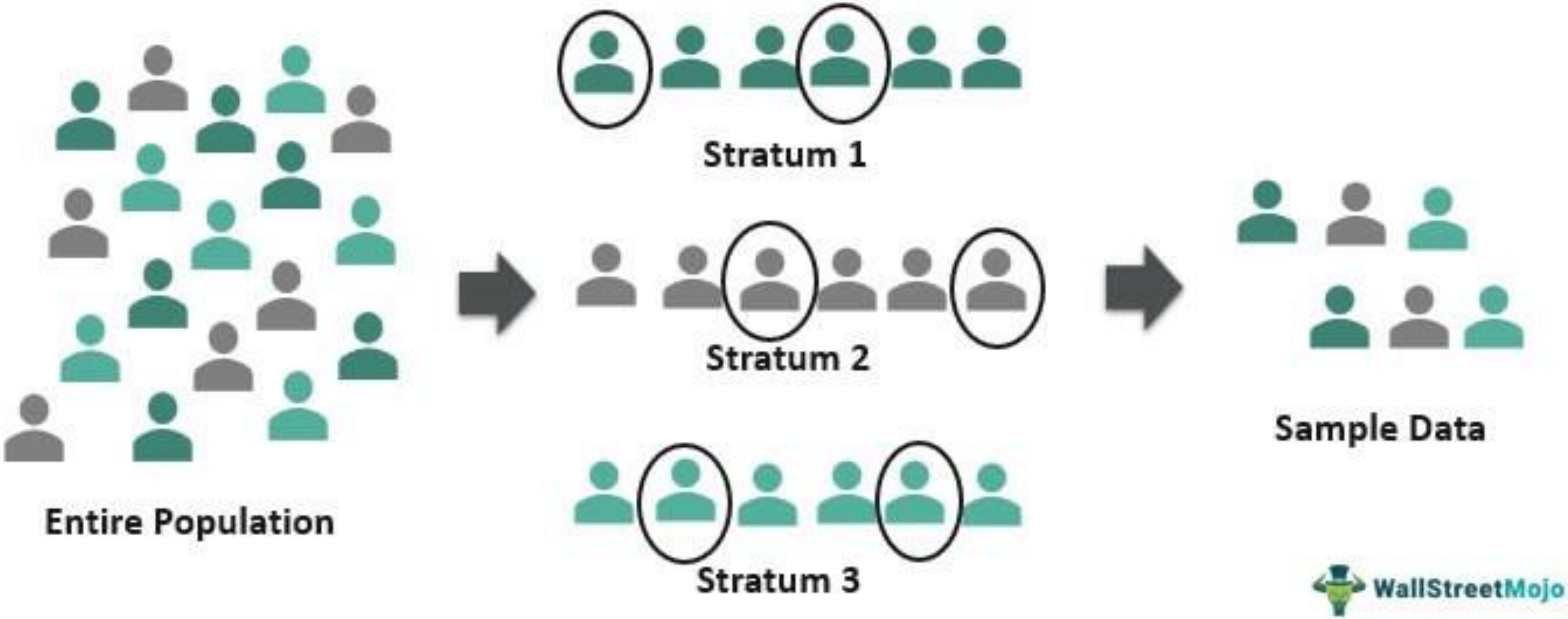
Structuring for Representation with Stratified Sampling



Divide the population into subgroups based on variables central to the analysis, then draw a simple random sample from each.

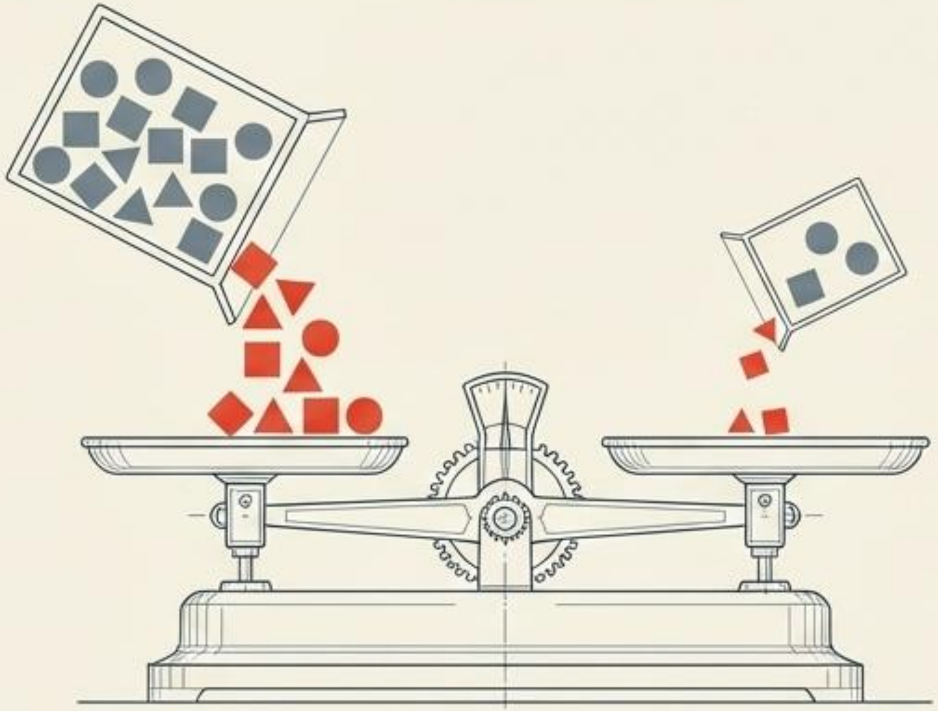
Stratified Sampling

Stratified sampling, refers to random sampling techniques that clubs items of whole population into different groups called strata, based on their similar characteristics. Then, samples from each stratum are taken, whether proportionately or disproportionately.



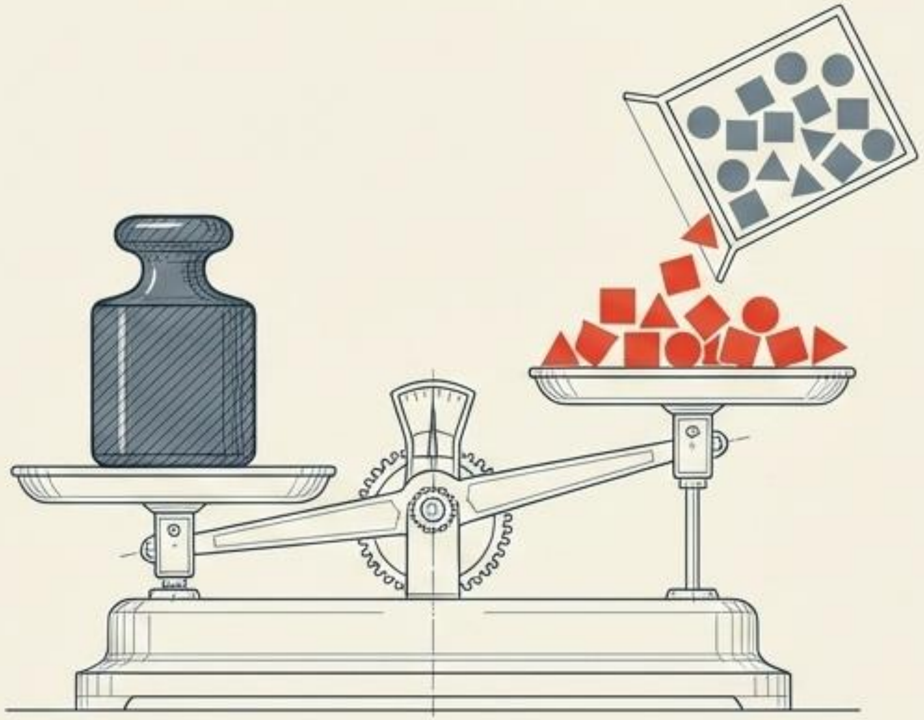
Balancing the Scales of Data

Proportionate Extraction



Sample size strictly reflects population size.
(Self-weighting).

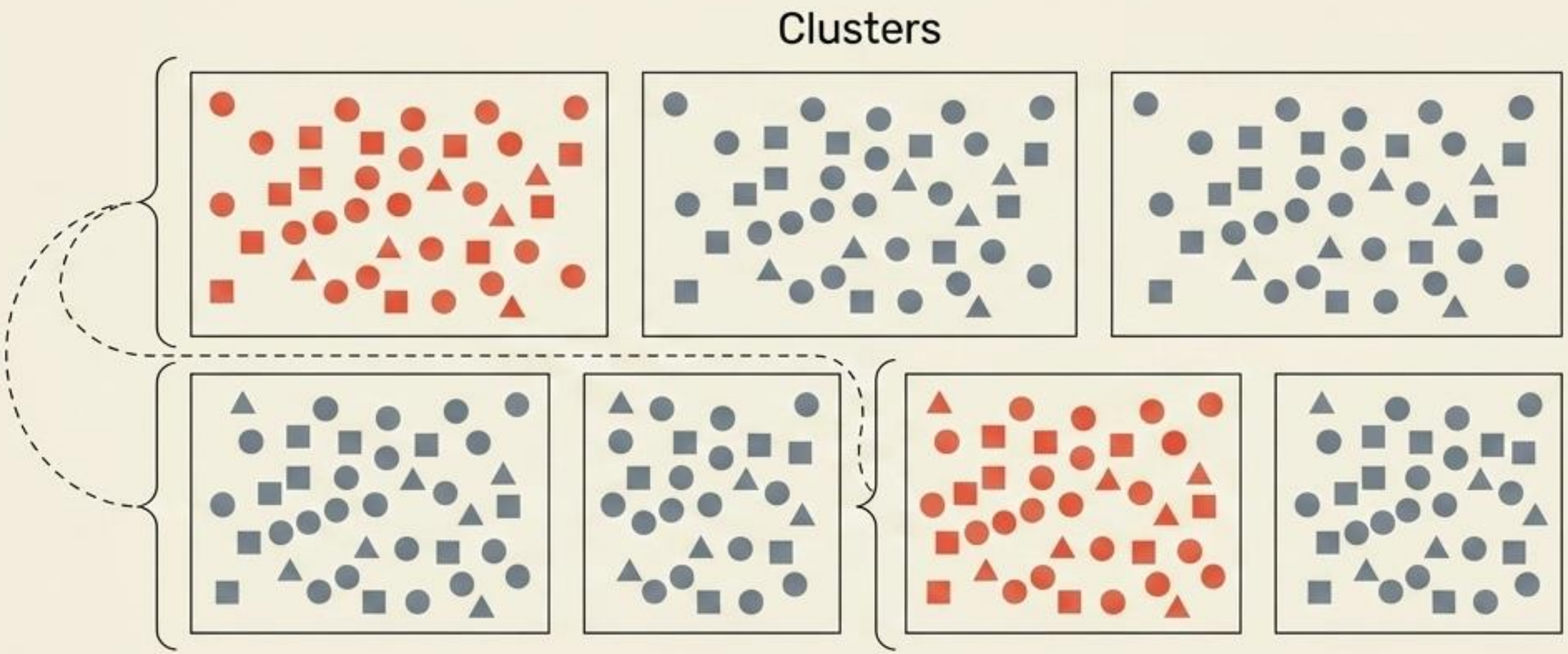
Disproportionate Extraction



Sample size does NOT reflect population size.
(Requires mathematical weights to ensure statistical significance).

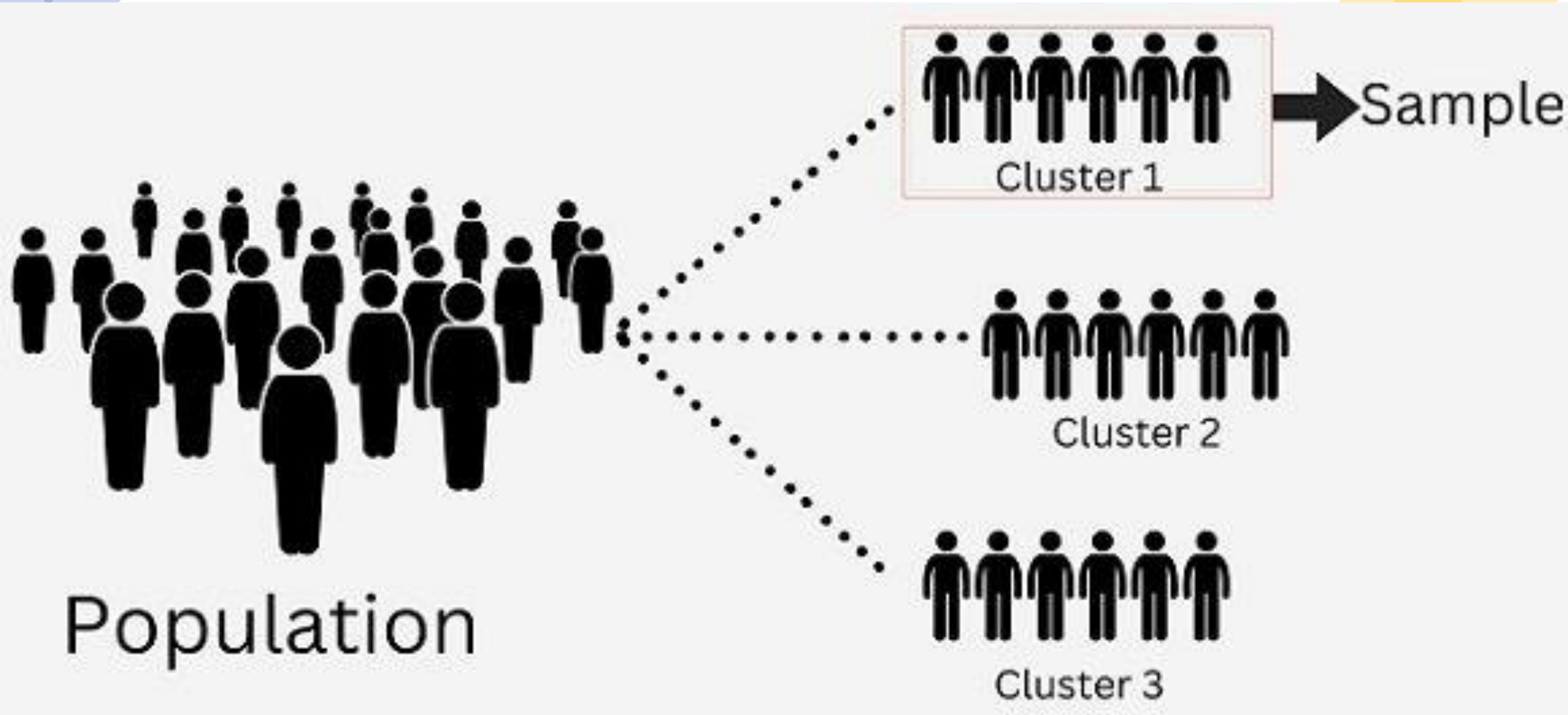
Cluster Random Sampling

Optimizing for Geographic and Logistical Cost

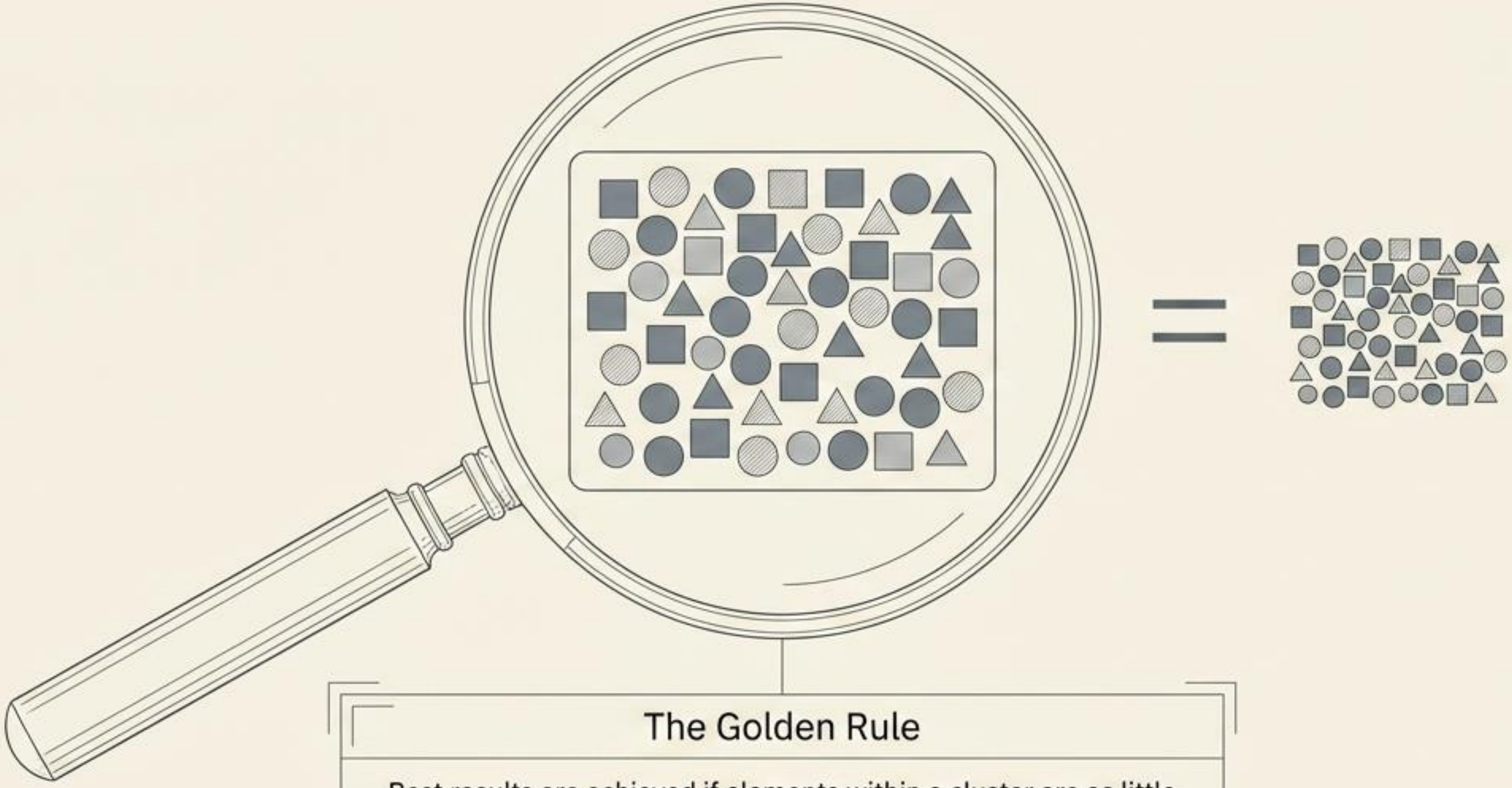


Divide population into Clusters.
Select clusters at random.
Take ALL elements within the chosen clusters.

Core Advantage:
Dramatically reduces the cost and travel time of sampling.


Sample
Cluster 1
Cluster 2
Cluster 3
Population

The Anatomy of a Perfect Cluster

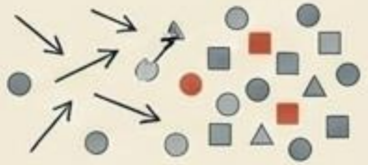
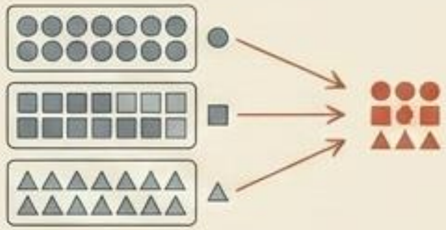
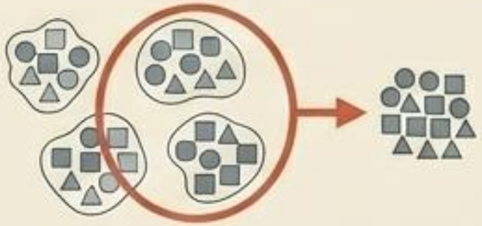


The Golden Rule

Best results are achieved if elements within a cluster are as little alike as possible. A cluster must represent a small-scale, internally diverse version of the entire population.

The Master Diagnostic Matrix

Save this matrix to instantly determine the correct filtration method for your dataset.

Method	Mechanism	Core Advantage	Visual Shorthand
Simple	Pure chance selection	Unbiased baseline	
Systematic	Every Kth member	Fast, linear execution	
Stratified	Subgroups by variable	Guarantees minority representation	
Cluster	Random whole groups	Dramatically reduces cost/logistics	

Types of Non-Random Sampling

The pragmatic alternative to pure probability.

The Definition

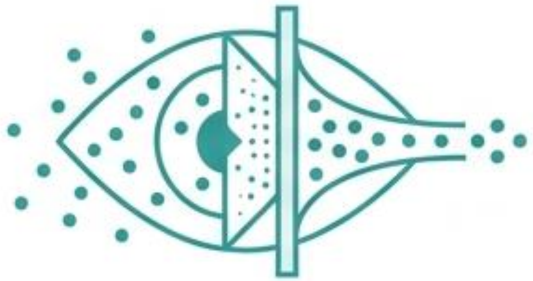
Non-random sampling (also known as non-probability sampling) abandons the requirement that every individual has an equal chance of selection. Instead, researchers rely on logistics, subjective expertise, or existing networks to build their sample.

The Tradeoff (Risk vs. Reward)

Four targeted tools for specific research scenarios

Convenience Sampling

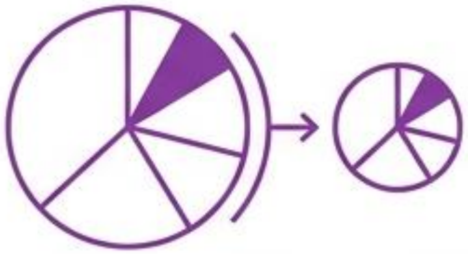
Selection by proximity.



Judgment Sampling

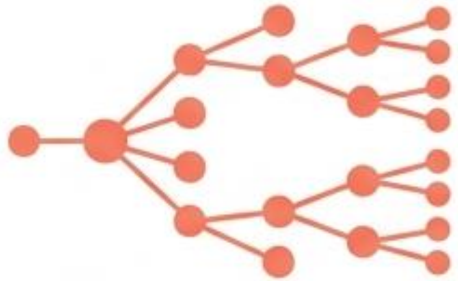
Selection by expert knowledge.

Crucial Distinction: These are NOT probability sampling methods. While often easier to perform, results cannot guarantee whole population representation and must be interpreted carefully.



Quota Sampling

Selection by demographic proportion.



Snowball Sampling

Selection by chain-referral.

Convenience Sampling

Convenience sampling relies on immediate subject accessibility

Proximity Map

Mechanic Summary

Subjects are chosen simply because they are the easiest to reach.

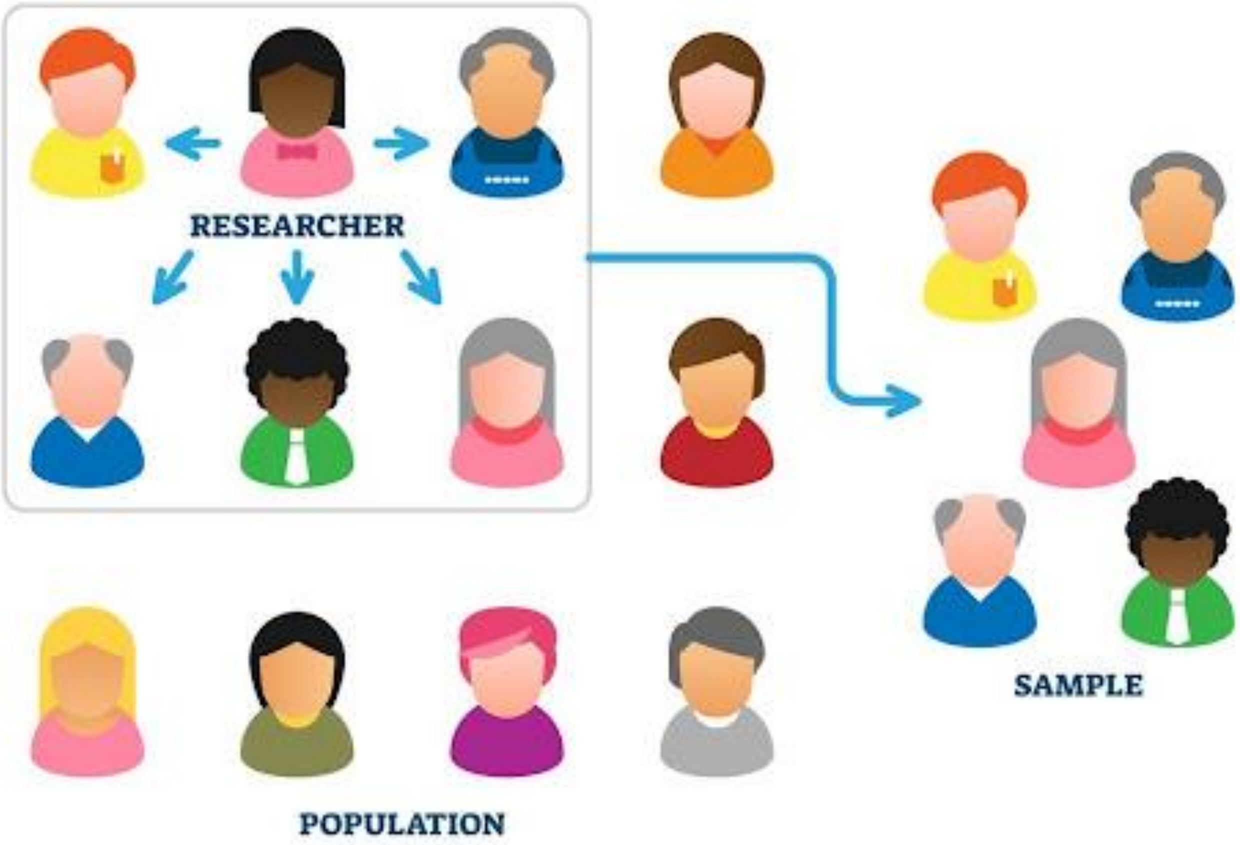
Primary Use Case

Often used for medical studies where voluntary participants are required.

Key Consideration

Easy to perform, but it cannot be guaranteed that voluntaries will represent the whole population.

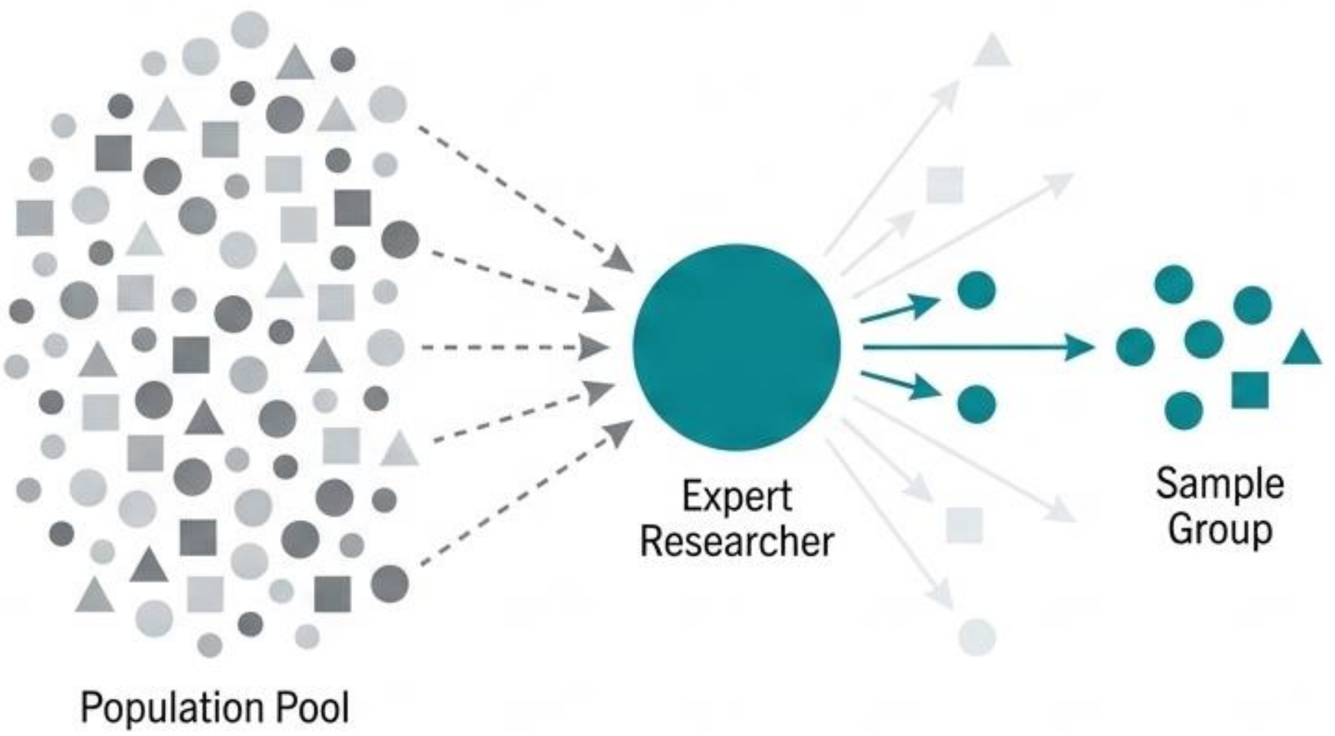
CONVENIENCE SAMPLING



Judgement Sampling

Judgment sampling deploys expert knowledge as a selection filter

Targeting Scope



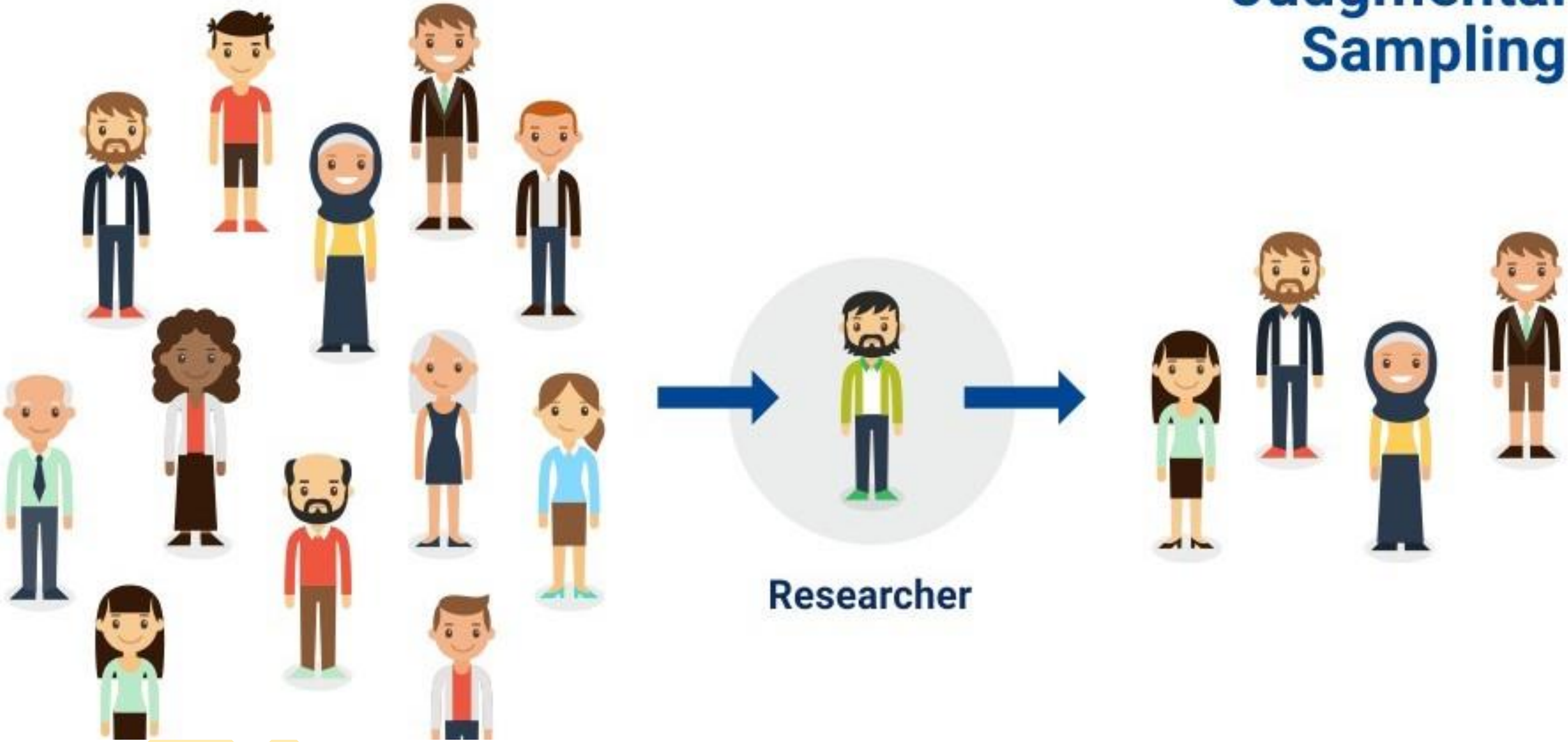
Mechanic Summary

The selection is entirely dependent on human expertise. Persons who are the most knowledgeable in the field explicitly select the specific items or subjects for the study.

Key Consideration

While easy to perform, the inherent reliance on human judgment means results should be interpreted and used carefully.

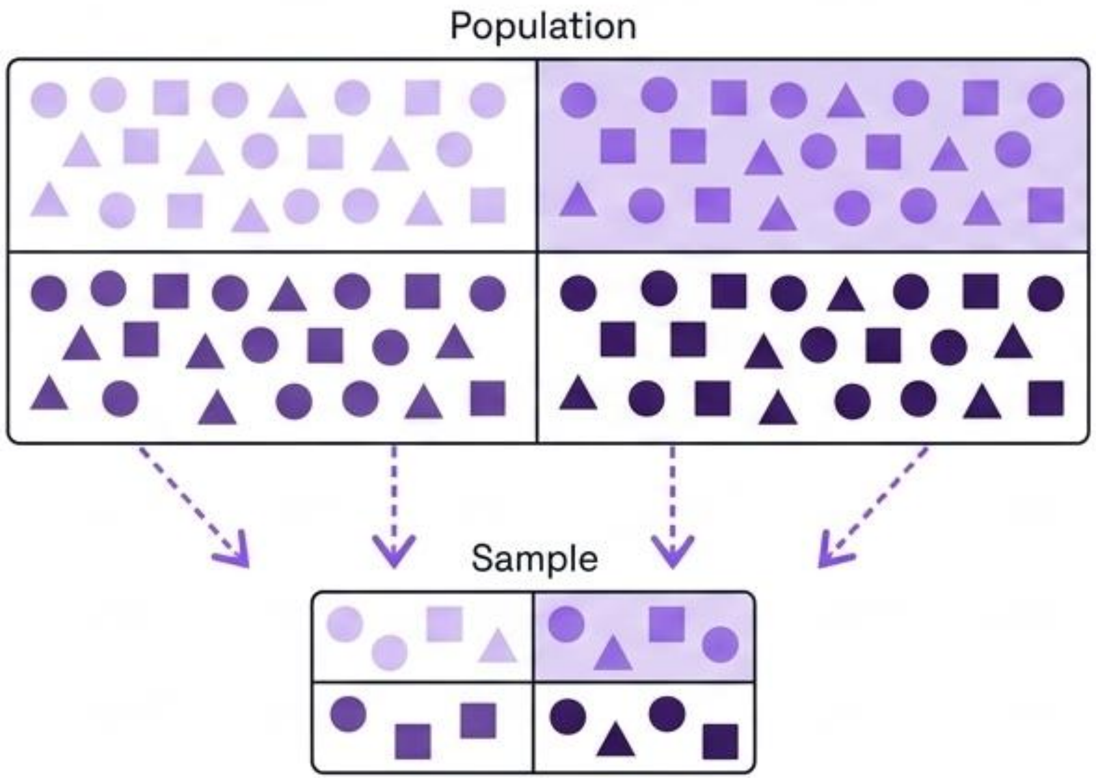
Judgmental Sampling



Quota Sampling

Quota sampling forces the sample to mirror specific population traits

Demographic Mirror



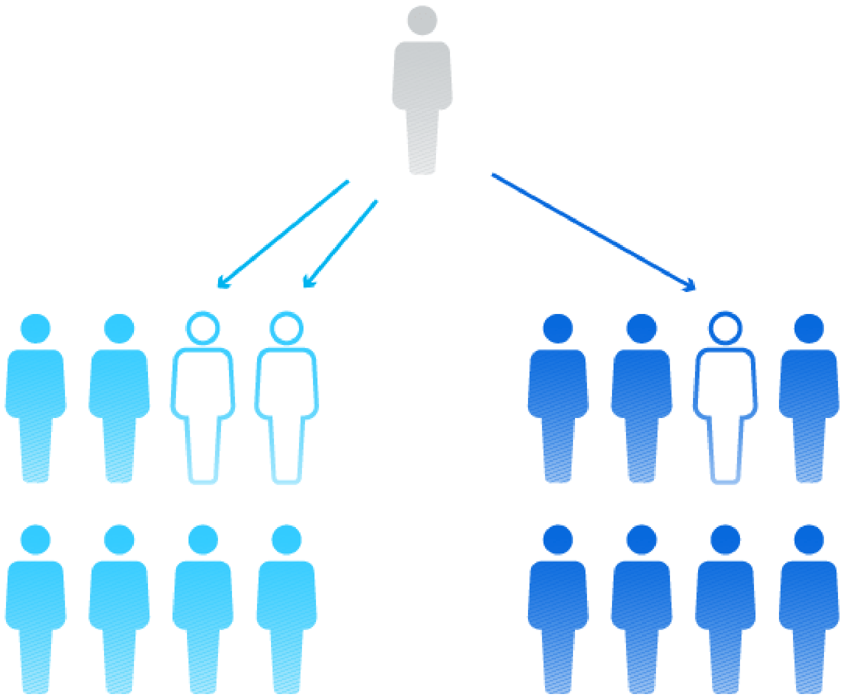
Mechanic Summary

The overarching rule of this method is **structural reflection:** the final makeup of the sample MUST **accurately reflect the makeup of the overall population** based on pre-selected characteristics.

Key Characteristics Tracked

Commonly used to structurally track demographic variables such as race, ethnic origin, and gender.

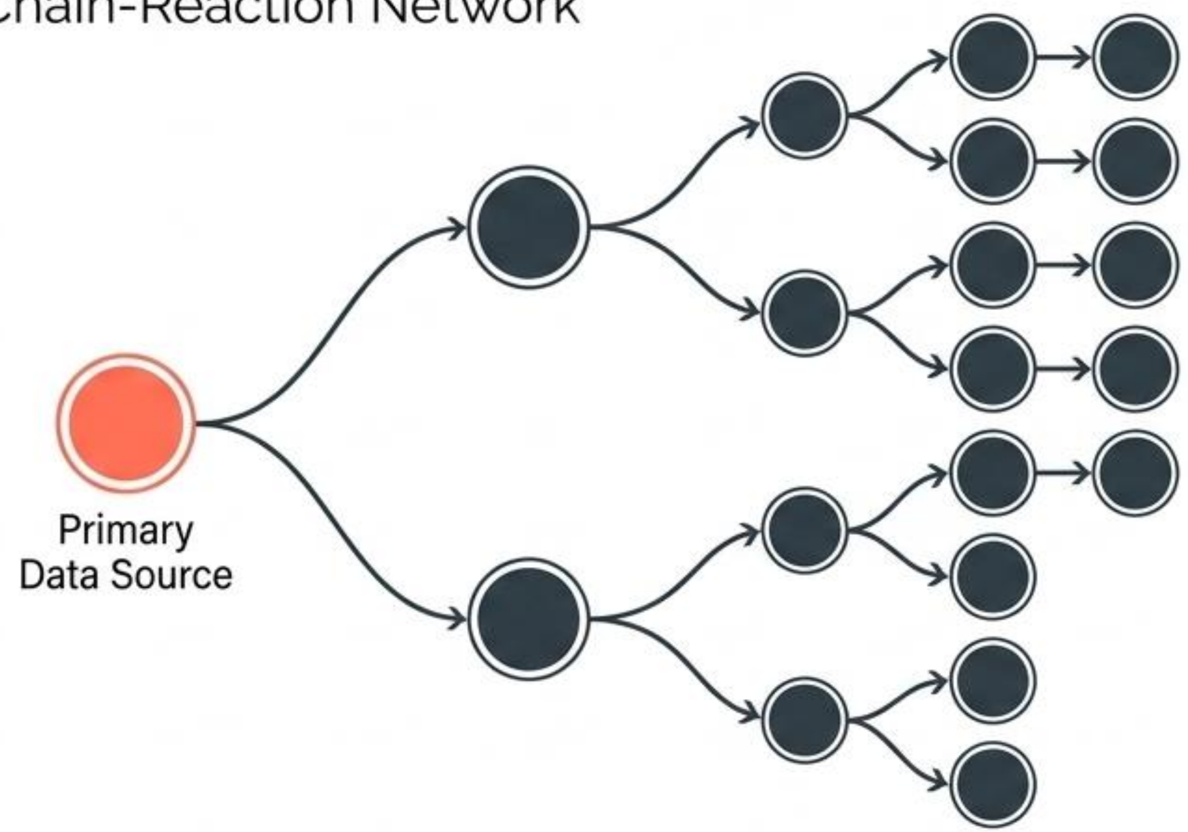
Quota sampling



Snowball Sampling

Snowball sampling leverages chain-referrals to uncover rare traits

Chain-Reaction Network



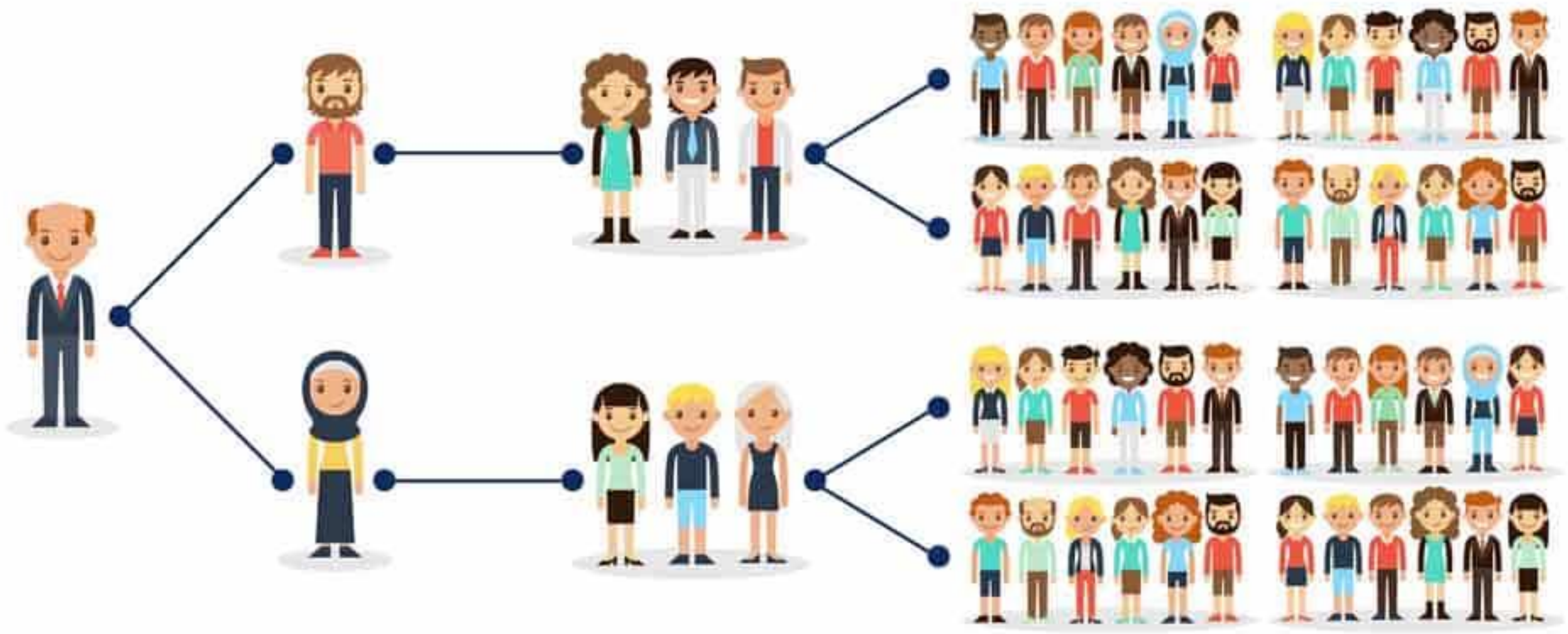
Mechanic Summary

Existing subjects provide referrals to recruit the required samples for the research study. A primary data source explicitly nominates other potential data sources who can participate.

Primary Use Case

This non-probability technique is specifically defined for scenarios where the target subjects possess rare, hard-to-find traits.

SNOWBALL SAMPLING



The Selection Diagnostic Matrix

Method	Core Mechanism	Ideal Application	Key Caveat
Convenience	Closest accessibility	Medical studies (volunteers)	No whole population guarantee
Judgment	Gatekeeper selection	Expert-driven field studies	Heavy reliance on human bias
Quota	Proportional mirroring	Tracking race, gender, ethnicity	Enforces structure but not randomness
Snowball	Chain-referral network	Populations with rare traits	Relies entirely on primary source networks